

The HTR Ecosystem — Complete Guide

Health Transformation Review

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The HTR Ecosystem — Complete Guide

User Guide · Technical Reference · Operations & Maintenance Manual

Health Transformation Review (HTR) / Health Transformation Solutions (HTS) *Covering the Book, the Platform, and the Academy*

Document version 1.0 — generated 2026-06-20 Applies to: Platform on Next.js 16.2 / React 19, Book manuscript v41, Academy (15 courses / ~90 tracks / ~243 lessons), Research Lab (24 tools)

How to read this guide

This guide serves three audiences, and each Part is self-contained so you can read only what you need:

- **Parts I–IV — Users.** Anyone using the book, the platform, the AI Analyst, or the Academy. No technical knowledge assumed. Written as step-by-step instruction.
- **Parts V–VI — Operators & developers.** Architecture, data model, the taxonomy spine, authentication, and the full environment reference.
- **Parts VII–VIII — Operations & launch.** The maintenance runbook, content operations, the AI corpus, monitoring, the pre-launch checklist, and the go-live runbook.

A reader who only wants to *use* HTR can stop after Part IV. A reader who *maintains or launches* it should read Parts V–VIII.

Throughout, a  marks a hands-on instruction, a  marks an operational hazard, and a  marks a file or code location.

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PART I — THE HTR ECOSYSTEM AT A GLANCE

1. What HTR Is and Why It Exists

Health Transformation Review (HTR) is an integrated knowledge and analytical system built to support the work of transforming a health system. It is not a single product but an ecosystem of three reinforcing components — a book, a web platform, and an academy — that together let a user learn a concept, see the evidence behind it, and model it against their own organization or state.

HTR uses **Vermont as its primary teaching case**. Vermont is unusually instructive because it is attempting comprehensive, all-payer health system transformation under hard statutory deadlines: Act 167 (2022) initiated the reform cascade; Act 68 (2025) made transformation mandatory; the AHEAD Model brings federal all-payer global budgets beginning January 2028; and a \$195M federal Rural Health Transformation (RHT) Program funds the infrastructure. Around this case, HTR layers **national comparative evidence** — Oregon’s Coordinated Care Organizations, California’s CalAIM, and all 50 states — so that Vermont’s specific lessons generalize to any state or system.

The reason the ecosystem exists at all is a single conviction, stated plainly in the book’s first chapter: **transformation fails when it is approached one pillar at a time**. Payment reform without clinical redesign produces savings that evaporate when care models do not change. Clinical improvement without payment reform produces better care that the payment system then penalizes. Technology investment without operational readiness produces platforms no one uses. The pillars must move together — and in the right order. HTR is built to teach, evidence, and model exactly that.

2. The Three Components and How They Connect

The Book — *Transforming American Healthcare: A Six-Pillar Framework for System Transformation*. Sixteen chapters plus front matter and appendices. It is the intellectual foundation: the framework, the dependency logic, the Vermont evidence, and the national comparisons. Everything else in the ecosystem is, in effect, an interactive expression of the book.

The Platform — a web application of 184 page routes. It contains six pillar hubs (one per pillar), a Research Lab of 24 interactive analytical tools, an AI Analyst, a 50-state dashboard, several standalone simulators, a daily intelligence feed (The Wire), and the full account/subscription system.

The Academy — the executive-education layer: 15 courses organized into roughly 90 tracks and 243 lessons, plus case studies, webinars, a faculty page, a glossary, getting-started guides, and credential paths.

These three are deliberately **cross-linked** so a user moves fluidly among learning, evidence, and modeling:

- **Book → Platform**. Each pillar chapter ends with a “Work This Chapter on the Platform” table that maps the chapter’s argument to the specific Research Lab tools that let the reader manipulate it. Reading about EHR feasibility in Chapter 4? The chapter table links you straight to the FHIR Lab, the EMR/EHR Lab, and the Statewide EHR Modeler.

- **Platform → Book.** Pillar hubs and the Research Lab carry “From the Book” callouts and chapter cross-links, so a user exploring a tool can jump to the chapter that explains the concept.
- **Academy → Platform.** Lessons carry “Apply It” callouts that deep-link to the matching tool — for example, the EHR business-case lesson links directly to the EMR/EHR Lab’s cost-modeling mode.
- **AI Analyst ↔ everything.** The Analyst is grounded in the book and Vermont source documents and knows the platform’s tool catalog, so it can answer a question *and* route the user to the right tool or page.

The mechanism that keeps all of this consistent is the **taxonomy spine** — three canonical lists (pillars, tools, chapters) in `frontend/lib/taxonomy/` against which every surface resolves. When those three lists agree, the ecosystem is aligned. (See §27.)

3. The Six-Pillar Framework

Every component of HTR is organized around the same six pillars. They are not arbitrary categories; they form a dependency structure.

Pillar	The question it answers	What it covers
Policy	Is it mandated?	Legislative architecture, Section 1115 waivers, CMMI models, the federal–state interface, budget neutrality
Economics	Does it pay?	Global budgets, reference-based pricing, alternative payment models, hospital finance, total cost of care
Technology	Can we measure it?	Data infrastructure, health information exchange, FHIR interoperability, EHRs, AI governance
Clinical	Does it improve care?	Care model redesign, risk stratification, quality measurement, care management
Equity	Does it close gaps?	Disparity measurement, social determinants of health, access, equity-weighted analysis
Operations	Is it executable?	Workforce, revenue cycle, supply chain, compliance, the administrative cost gap

The dependency logic. Technology and Policy are upstream *gates*. Economics, Clinical, and Equity depend on them — you cannot run a global budget (Economics) for a population you cannot measure (Technology), and you cannot enforce a payment model (Economics) without a policy mandate (Policy). Operations is the execution layer that translates all of the above into organizational reality. The framework’s central, testable claim is that **funding a downstream pillar before its upstream gate is open produces a measurable failure cascade** — the exact dynamic that sank earlier reform efforts, and the principle a user can reproduce hands-on in the HTR Simulator.

4. Who HTR Is For

HTR is built for the people who make or execute transformation decisions:

- **Health-system executives** (CEOs, CFOs, CMOs, CMIOs) navigating value-based care, technology adoption, and the move to global budgets.

- **Policy analysts and government officials** designing waivers, CMMI applications, and state reform legislation.
- **Health economists and actuaries** modeling payment models, rates, and cost-effectiveness.
- **Clinicians and clinical informaticists** redesigning care and managing risk.
- **Health-technology professionals and investors** evaluating interoperability, AI, and digital health.
- **Students and researchers** studying health system transformation.

The Academy's getting-started flow lets a user declare their role, which personalizes what the platform and AI Analyst surface first.

PART II — THE BOOK

5. Overview and Structure

Title: *Transforming American Healthcare: A Six-Pillar Framework for System Transformation*

Version: v41 **Length:** 16 chapters, plus a Preface, an Introduction, and a set of appendices.

The chapters are grouped by pillar, with most pillars getting a “pillar” chapter (the architecture/theory) followed by a “pillar in practice” chapter (implementation):

- **Chapter 1** — The Six-Pillar Framework and the Execution Sequence. The dependency logic and the failure-cascade argument.
- **Chapters 2–3** — The Policy Pillar: legislative architecture for structural reform; then CMMI models, waiver strategy, and the federal–state interface.
- **Chapters 4–5** — The Technology Pillar: data infrastructure (VHCURES, VITL/VHIE, the statewide-EHR feasibility question); then FHIR implementation, AI governance, and clinical decision support.
- **Chapters 6–7** — The Economics Pillar: global budgets, reference-based pricing, financial reform; then VBC financial modeling and APM readiness.
- **Chapters 8–9** — The Clinical Pillar: care delivery redesign; then care-model implementation and quality mechanics.
- **Chapter 10** — The Equity Pillar: closing gaps rather than averaging them.
- **Chapter 11** — The Operations Pillar: executing hospital system transformation, including the administrative cost gap.
- **Chapters 12–16** — Infrastructure for knowledge transfer; the future of transformation; political sustainability across election cycles; healthcare transformation as portfolio management (PMI standards); and the AHS restructuring roadmap.

The appendices include **Appendix E — HTR Platform Tools and Methodology Overview**, which catalogs all 24 Research Lab tools and is the book’s canonical bridge to the platform.

6. The Source Files and the Sync Rule

The book exists in three forms; understanding which is authoritative is essential for maintenance.

📁 **HTR_Book_v41.docx** — the editable manuscript. This is what the author round-trips through Google Docs / Microsoft Word. It is **formatting-sensitive**. 📁 **HTR_Book_v41.md** — a Markdown twin of the .docx, kept in sync with it. 📁 **HTR_Book_v41.pdf** — a derived, distributable output.

⚠️ **The sync rule (critical for maintenance):** when the book changes, edit **both** the .docx and the .md, with the *same* change, so they never drift apart. The .docx must be edited carefully because its formatting is fragile (it is round-tripped through word processors). The .md mirrors it. Never edit one and leave the other stale — that creates silent divergence between what’s published and what the platform/tools reference. The .pdf is regenerated from these when a new distributable is needed.

⚠️ The .docx files are **gitignored** (only the .md is in version control), so the .docx must be backed up separately.

A further distinction that matters: **“the book”** means the manuscript (the .docx/.md). **“The /book page”** means the frontend code that renders the book browser on the platform. They are different things; a change to one is not automatically a change to the other.

7. Reading the Book on the Platform

The book is fully integrated into the platform, not merely linked as a download:

-  **/book** — the chapter browser. It lists every chapter with a description, the key concepts introduced, reader-profile guidance (“if you are a hospital executive...”), and the per-chapter platform cross-links. Start here.
-  **/read/[slug]** — the in-app reader for a single chapter. Supports **bookmarking** (save your place / mark a chapter) and **chapter notes** (your own annotations, stored to your account).
-  **/book/listen** — the audio narration player.

The book is also surfaced prominently across the platform: a hero slide on the home page, a pinned entry at the top of the Home Sidebar (with a “New” badge), and a  **THE BOOK** link in the header’s top bar and the Learn mega-menu.

8. Audio Narration

Every chapter has full audio narration so the book can be consumed hands-free.

 Audio files (.m4a) and their transcripts (.txt) live in frontend/public/audio/narration/, named in reading order: 00-preface, 01-introduction, 02-chapter-01, through the final chapter. The /read/[slug] reader displays the transcript; /book/listen plays the audio with chapter navigation.

 **Operational note:** the narration audio totals roughly **280 MB**. It is served as static CDN assets and is deliberately **excluded from the serverless function bundle** via `outputFileTracingExcludes` in `next.config.ts`. This exclusion is what keeps the /read/[slug] deployment under Vercel’s 300 MB function-size limit (the route reads the small .txt transcripts at runtime, which would otherwise drag all the audio into the function). Do not remove or weaken that exclusion without understanding this constraint (see §31).

9. “Work This Chapter on the Platform”

This is the book’s signature integration feature and the clearest expression of the ecosystem idea. Each pillar chapter ends with a section titled **“Work This Chapter on the Platform,”** containing a short intro and a table of this form:

Do this	On this tool	What to look for
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Each row maps one of the chapter’s arguments to a specific Research Lab tool and tells the reader what insight to look for. For example, Chapter 4 (Technology Pillar) includes rows for the FHIR Interoperability Lab, the Risk Stratification Engine, the EMR/EHR Lab, and the Statewide EHR Deployment Modeler. Chapter 11 (Operations) includes the Transformation Scorecard, the Workforce Modeler, and the Vermont-RHT CIN & Shared Services and EMS Transformation modelers.

This is how a reader moves from *understanding* a concept to *modeling it against their own numbers*. It is also a maintenance obligation: when a tool is added to the platform, the relevant chapter's table — and Appendix E — must be updated to match (see §34).

10. Appendix E and Book↔Platform Alignment

Appendix E is the book's canonical list of the platform's tools. It states the tool count and enumerates the tools by bench. Because it is canonical, it is the place most likely to go stale when the platform grows, and keeping it current is part of the alignment discipline.

“Alignment” across the ecosystem means three things agree: 1. The **tool registry** (`tools.ts`) — what tools exist. 2. The **chapter tables + Appendix E** in the book — which tools the book references. 3. The **Academy** — where lessons point users to tools.

When all three agree, a reader who learns a concept in the book, looks it up in the Academy, and opens the tool on the platform finds a consistent story. The new-tool checklist in §34 exists precisely to preserve this alignment.

PART III — THE PLATFORM

11. Navigation and Layout

The platform uses a **sidebar-first** navigation model with three persistent regions plus the main content area.

Home Sidebar (left) — the primary navigation. From top to bottom: - **The Book**, pinned at the top. - **The six pillars** (Policy, Economics, Technology, Clinical, Equity, Operations), each expandable to its intelligence pages and its Research Lab tools. - **The Research Lab**, grouped by pillar. - **The Academy**. - **Dashboards and simulators**.

Sidebar accordions support multiple sections open at once. The sidebar’s tool listings are driven by `labToolIds` arrays in `frontend/components/HomeSidebar.tsx`, which resolve against the tool registry — so a tool only appears in the sidebar if its ID is listed there (a common omission when adding tools; see §34).

Header (top) — the brand, the “ THE BOOK” link, a “Learn” mega-menu (curated links to pillars, the Research Lab, the Academy, and reference material), and account/login controls.

AI Analyst (right sidebar) — a persistent assistant. Collapsed, it’s a compact widget; expanded, it becomes a chat panel; or open it full-screen at `/chat`.

Main content (center) — the active page.

The layout is fully responsive: on mobile the sidebars collapse into accordions and the AI Analyst becomes a launchable panel.

12. The Six Pillar Hubs

Each pillar has a hub page — `/policy`, `/economics`, `/technology`, `/clinical`, `/equity`, `/operations` — generated from one shared template (`PillarOverview`) with per-pillar content from `frontend/lib/data/pillar-topics.ts`. Every hub contains, in order:

1. **A hero** with the pillar’s eyebrow, title, and tagline.
2. **A “From the Book” callout** linking to the pillar’s chapter.
3. **A topic-card grid** — the sub-topics within the pillar (e.g., Technology’s cards: AI & Machine Learning, Digital Health & Telemedicine, Data Security & Governance, Tech-Enabled Workflow). Each card links to a dedicated intelligence page.
4. **A “Tools & Data” footer** — the Research Lab tools relevant to that pillar, plus a link to the full lab.
5. **Latest related editorial / analysis**.

 **To explore a pillar:** open its hub from the sidebar, read the From-the-Book callout for grounding, browse the topic cards for the sub-area you care about, then jump to the linked tools in the footer to model it.

13. The Research Lab — Overview

The Research Lab (`/research-lab`) is the analytical core of the platform: **24 interactive tools** that are self-contained, browser-based, and run entirely client-side. They perform real computation (Monte Carlo simulations, Markov models, FHIR resource generation, financial projections)

using only inputs you provide through the UI — **no data upload, no PHI, no external dependencies**. Every tool’s figures are illustrative planning models, clearly labeled as such, not audited or live data.

The tools are organized into **seven benches**, each its own route:

Bench	Route	Tools
Interoperability & Risk	/research-lab/interoperability	FHIR Lab, Risk Stratification Engine, EMR/EHR Lab, Statewide EHR Modeler
Payment Models & VBC	/research-lab/payment-models	APM Design Lab, Shared Savings Calculator, CEA Calculator, Global Budget Transition Modeler
Policy & Quality Sciences	/research-lab/policy-quality	Policy Simulator, Clinical Quality Optimizer, Hospital Financial Stress Test, HTA Studio, Actuarial Lab, Work Requirements Calculator, H.R. 1 Cliff Scenario
Population & Equity	/research-lab/population-equity	Population Health Modeler, Health Equity Studio
Technology & AI	/research-lab/technology-ai	AI Clinical Governance Lab, Digital Health Lab
VBC, Clinical & Quality	/research-lab/vbc-clinical-quality	Clinical Data Exchange Lab, Risk Stratification Methodology, VBC Quality Measures, High vs. Low Value Care
Knowledge & Workspace	/research-lab/knowledge-workspace	Transformation Scorecard, VBC Readiness Assessment, Evidence Library, Workforce Modeler, Innovation Leaderboard, Research Workspace, CIN & Shared Services Modeler, EMS Transformation Modeler

 **Using any tool — the universal pattern:** 1. Open the bench route and click the tool’s tab. 2. Read the tool header (it states the tool’s purpose and methodology). 3. Adjust the **input controls** on the left/top — sliders, dropdowns, toggles, number fields. 4. Read the **outputs** — stat cards, tables, charts, and (for several tools) a plain-language verdict. 5. Ask the **AI Analyst** to interpret the result. 6. **Bookmark** the tool for return visits.

Deep-linking: every tool is addressable via ?tab=<id> on its bench. Some tools additionally support &mode=<id> to open a specific sub-mode (e.g., /research-lab/interoperability?tab=emr&mode=workflow opens the EMR/EHR Lab directly in its Workflow Simulator). Academy “Apply It” callouts use these deep links.

The canonical in-app usage guide — who each tool is for, when to use it, one power tip each — is at /academy/getting-started/research-lab.

14. The Research Lab — Tool-by-Tool Walkthroughs

This section documents each tool in depth: who it is for, when to use it, the inputs it takes, the outputs it produces, and the methodology behind it. Tools are grouped by bench. Every tool is a self-contained client-side model; all figures are illustrative planning outputs, not live or audited data.

Bench 1 — Interoperability & Risk

FHIR Interoperability Lab — ?tab=fhir

For: health IT professionals, clinical informaticists, integration engineers. **When:** planning a Patient Access or Provider Directory API, testing FHIR R4 resource design, mapping terminologies, or preparing for an ONC information-blocking review. **Inputs:** resource type and parameters; terminology terms to map; a CDS Hooks scenario; a prior-authorization case. **Outputs:** spec-compliant FHIR R4 JSON (with generated UUIDs, meta.profile, narrative), cross-terminology mappings (ICD-10 ↔ SNOMED ↔ LOINC ↔ RxNorm), a simulated CDS Hooks exchange, a CMS prior-auth workflow trace, and an ONC compliance checklist. **Methodology:** the builder generates genuine R4 structures; terminology mappings draw on standard code systems. **Power tip:** start from the rule or use case, not the resource — the lab walks you to the relevant resources and checkpoints.

Risk Stratification Engine — ?tab=risk

For: population-health managers, data scientists, care-management leaders. **When:** building a risk-stratified care-management program, validating HCC scores, or designing the data inputs for a predictive model. **Inputs:** a population's diagnosis profile; the risk model to apply; comorbidity selections. **Outputs:** HCC v28 RAF scores, a risk-tier segmentation, custom-model results, and comorbidity-interaction analysis using Elixhauser and Charlson indices. **Methodology:** HCC v28 RAF logic with standard comorbidity indices. **Power tip:** run the RAF module first to understand the population's risk profile before designing interventions.

EMR/EHR Lab — ?tab=emr

For: CMIOs, health-IT leaders, hospital finance teams. **When:** selecting an EHR, building the adoption business case, auditing data quality, or quantifying documentation burden. **Structure:** an integrated four-mode tool driven by one vendor selection — choose a vendor and that choice flows through every mode. - *Vendor Comparison* (&mode=vendors): weight interoperability, usability, cost, and ambulatory fit; rank Epic, Oracle Health, MEDITECH, athenahealth, with public market-share context. - *Adoption & Cost* (&mode=cost): inputs — providers, timeline, hosting model, clinical revenue/provider. Outputs — implementation capex, annual license, go-live productivity loss, a 5-year cost/benefit table, and a break-even year. - *Data Quality* (&mode=quality): audits a mock record against the ten USCDI classes, seeded by the selected vendor's interoperability score. - *Workflow Simulator* (&mode=workflow): models a clinician's daily EHR minutes by vendor and panel; toggle scribes, ambient AI, templates, and team-based care against the ~360 min/day national benchmark. **Power tip:** the whole tool reflects one vendor end-to-end — change the vendor and re-read all four modes to compare options. See the full worked example in §41.1.

Statewide EHR Deployment Modeler — ?tab=statewide-ehr

For: state health officials, CMIOs, policy analysts (the Act 167 / Oliver Wyman feasibility question). **When:** weighing a single statewide EHR against FHIR interoperability across existing platforms. **Inputs:** participating hospitals; per-hospital migration vs. integration costs; data-timeliness and disruption assumptions. **Outputs:** two paths side by side — *single statewide EHR* vs. *FHIR interoperability* — each scored on 10-year total cost, data timeliness, migration disruption, and vendor lock-in, with a verdict. **Methodology:** a structured cost/benefit comparison anchored to Vermont's feasibility-assessment framing. **Power tip:** the central question is whether

FHIR delivers acceptable real-time exchange; if it does, the case for a disruptive migration weakens.

Bench 2 — Payment Models & VBC

APM Design Lab — ?tab=apm-design

For: health economists, payment-model consultants, payers, hospital finance. **When:** designing a new APM, evaluating a structure, or preparing a CMMI application. **Inputs:** model type, risk arrangement, benchmark parameters. **Outputs:** episode bundles, global-budget structures, benchmark waterfall charts, and natural-language model recommendations. **Power tip:** start with model type, then configure the risk arrangement; use the recommendation engine if unsure.

APM Shared Savings Calculator — ?tab=apm-calc

For: ACO finance teams, actuaries. **When:** projecting shared savings/losses under MSSP, ACO REACH, or a custom global budget. **Inputs:** benchmark methodology, spending, risk-corridor and quality-withhold parameters. **Outputs:** projected savings/losses, risk-corridor effects, quality-withhold impact.

CEA Calculator — ?tab=cea

For: health economists, HTA analysts. **When:** assessing an intervention's value. **Inputs:** cost and effectiveness estimates, time horizon, willingness-to-pay threshold. **Outputs:** cost per QALY, NNT, break-even timeline, and comparison against ICER/NICE/CMS thresholds.

Global Budget Transition Modeler — ?tab=gb-transition

For: hospital finance, state regulators. **When:** modeling a move from fee-for-service to a global hospital budget. **Inputs:** baseline volumes/revenue, transition timeline, population. **Outputs:** a multi-year transition trajectory and the financial inflection points.

Bench 3 — Policy & Quality Sciences

Policy Simulator — ?tab=policy

For: policy analysts, government officials. **When:** modeling 1115 waivers, global-budget designs, Medicaid expansion, or price-transparency policy. **Inputs:** waiver type, state scenario, policy levers. **Outputs:** modeled impact on utilization, cost, and coverage against pre-loaded state scenarios.

Clinical Quality Optimizer — ?tab=quality

For: quality officers, VBC teams. **Inputs:** measure selections, current performance, target. **Outputs:** HEDIS simulation with NCQA benchmarks, CMS Star prediction, MIPS optimization, P4P ROI.

Hospital Financial Stress Test — ?tab=scorecard

For: hospital CFOs, boards, state regulators. **When:** stress-testing financials against payer-mix shifts, Medicaid rate cuts, and volume changes. **Inputs:** peer archetype (CAH, Rural PPS, Urban Tertiary), stressors. **Outputs:** a 10-year margin trajectory. See worked example §41.6.

HTA Studio — ?tab=hta

For: HTA analysts, coverage decision-makers. **Outputs:** budget-impact models, multi-criteria decision analysis (8 criteria), and real Monte Carlo probabilistic sensitivity analysis (1,000 iterations across Beta/Log-normal/Gamma distributions).

Actuarial Lab — ?tab=actuarial

For: actuaries, plan finance. **Outputs:** ACA actuarial value, premium development (3 methodologies), adverse-selection (“death spiral”) modeling, IRA drug-pricing impact.

Work Requirements Calculator — ?tab=medicaid-wr

For: policy analysts, Medicaid officials. **Outputs:** enrollment/disenrollment under work-requirement policies, including the gap between nominal exemptions and real-world coverage loss from administrative churn.

H.R. 1 Cliff Scenario — ?tab=hr1-cliff

For: state budget officials, hospital finance. **Outputs:** the financial cliff for states and providers under H.R. 1 Medicaid provisions.

Bench 4 — Population & Equity

Population Health Modeler — ?tab=population

For: population-health managers, public-health researchers. **Outputs:** Markov disease-progression models (5 conditions), SIR epidemic dynamics, preventable-hospitalization modeling, intervention ROI.

Health Equity Studio — ?tab=equity

For: equity officers, public-health researchers. **Outputs:** disparity analysis across 10 outcomes by race/ethnicity/income/geography, geographic-access mapping, SDOH burden scoring, and the HEROI equity-weighted ICER.

Bench 5 — Technology & AI

AI Clinical Governance Lab — ?tab=ai

For: AI governance leads, clinical informaticists. **When:** building an AI governance framework, evaluating a clinical AI tool, or preparing for FDA SaMD / ONC HTI review. **Outputs:** model-performance comparison, bias detection (demographic parity, equal opportunity), governance frameworks, AI ROI.

Digital Health Lab — ?tab=digital

For: digital-health product teams, investors. **Outputs:** RPM ROI using CMS CPT codes (99453–99458), telehealth-utilization scenarios, patient-engagement platform comparison, EHR-interoperability optimization.

Bench 6 — VBC, Clinical & Quality

Clinical Data Exchange Lab — ?tab=hl7

For: integration engineers, informaticists. **Outputs:** annotated HL7 v2 messages (ADT/ORU), FHIR R4 bundles, an HL7↔FHIR side-by-side bridge, and a USCDI v3 browser — anchored to 8 Vermont patient scenarios.

Risk Stratification Methodology — ?tab=risk

For: data scientists, actuaries. **Outputs:** step-by-step HCC v28 RAF calculation, a population-tier pyramid, and comparison of ACG, CDPS, and Charlson algorithms.

VBC Quality Measures — ?tab=quality

For: quality teams. **Outputs:** HEDIS measures with numerator/denominator logic, 30-day readmission analysis (CMS RSRR), AHRQ Prevention Quality Indicator avoidable-ED classification.

High vs. Low Value Care — ?tab=value

For: clinical leaders, VBC teams. **Outputs:** A1C/BP panel management with shared-savings math, a Choosing Wisely low-value-care scan, and total-cost-of-care decomposition by service category.

Bench 7 — Knowledge & Workspace

Transformation Scorecard — ?tab=scorecard

For: executives. **Outputs:** a six-pillar executive scorecard with Vermont AHEAD statutory milestones integrated.

VBC Readiness Assessment — ?tab=readiness

For: transformation leaders. **Outputs:** a 30-dimension, 6-domain readiness score with prioritized gap analysis; presets for Vermont AHEAD, CAH, and advanced systems.

Evidence Library — ?tab=evidence

For: researchers, all users. **Outputs:** searchable landmark CEA/CUA studies, CMMI models with lessons learned, and HTR policy briefs.

Workforce Modeler — ?tab=workforce

For: workforce planners, operations. **Outputs:** physician/nurse supply-demand projections across specialties over 10 years, staffing-ratio impacts, turnover costs, rural-incentive modeling.

Innovation Leaderboard — ?tab=Leaderboard

For: benchmarking, strategy. **Outputs:** rankings of states, health systems, and payers on transformation and VBC-maturity indices.

Research Workspace — ?tab=workspace

For: all users running extended analyses. **Outputs:** saved/compared scenarios, structured reports from templates, citation management (AMA/APA), and Markdown/text export.

CIN & Shared Services Modeler — ?tab=cin (Vermont RHT)

For: hospital finance, system operators. **Inputs:** hospitals, discharges, the administrative cost premium (~\$1,303/discharge), shared-service domains, group-purchasing volume. **Outputs:** annual admin savings, group-purchasing savings, CIN setup/operating cost, and break-even in months. See worked example §41.3.

EMS Transformation Modeler — ?tab=ems (Vermont RHT)

For: EMS planners, state operations. **Inputs:** EMS agencies (31), target regions, 911 volume, non-emergent share, community-paramedicine reach. **Outputs:** ED visits diverted, admissions prevented, regionalization savings, program cost, net annual impact. See worked example §41.4.

Top-level simulators (not on a bench)

These are standalone routes, not Research Lab tabs: - **HTR Simulator** (/htr-simulator) — score an organization or state across all six pillars and watch the dependency cascade (worked example §41.5). - **Transformation Friction Index** (/transformation-friction-index) — identify the binding-constraint pillar. - **Impact Simulation** (/impact-simulation) — model intervention impact. - **HTI Dashboard** (/hti-dashboard) — the composite Health Transformation Index. - **Investment Tracker** (/investment-tracker) — capital flows and digital-health funding. - **Medicaid Eligibility Simulator** (/medicaid-eligibility-simulator) — eligibility scenarios.

15. The AI Analyst

The AI Analyst is a retrieval-augmented assistant, available as a right-sidebar widget on every page and full-screen at /chat.

What it can do: - Answer questions grounded in the book and Vermont source documents (it retrieves from a curated corpus rather than answering from generic training). - Recommend and link to the right Research Lab tool or platform page for a question (e.g., “how do I compare EHR vendors?” → the EMR/EHR Lab). - Help interpret a tool’s output.

How it works (user-relevant): the Analyst is *retrieval-augmented* — it searches an indexed corpus (the book, Vermont policy documents, the Research Lab documentation) and a structured catalog of every platform page and tool, then composes an answer grounded in what it found. This is why its tool recommendations are accurate and current — provided the catalog and corpus are kept up to date (see §35).

Access: the widget is available broadly; full chat is gated to subscribers and above. The frontend /api/chat endpoint is a thin proxy to the Python AI backend.

 **Tips:** ask it your real question in plain language; ask it “which tool should I use to...”; after running a tool, paste your result and ask it to interpret. If it ever cannot reach the backend, it returns a clear error rather than a wrong answer.

16. Dashboards, Simulators, and The Wire

- **50-State Dashboard** (/dashboard, /dashboard/[state], down to /dashboard/[state]/hospitals/[hospital]) — hospital-level and state-level transformation data, including RHT participation and financial metrics.

- **HTI Dashboard** (/hti-dashboard) — the composite Health Transformation Index across states/systems.
- **HTR Simulator** (/htr-simulator) — the flagship six-pillar dependency simulator.
- **State comparisons & spotlights** — /compare-states, /states/[state], plus dedicated Vermont, Oregon CCO, and California CalAIM pages.
- **The Wire** (/the-wire) — a daily, RSS-driven intelligence feed across all six pillars, with reader comments.

Detail — how a free visitor experiences the platform. Without an account, a visitor can read every pillar hub, browse the book at /book, read The Wire, see pricing, and use the AI Analyst widget in a limited capacity. This is the public “front door” and is what most launch traffic will see. The conversion path is deliberate: a free visitor exploring a pillar hub encounters “From the Book” callouts and tool links; clicking a gated tool (e.g., the full AI chat) routes them to /upgrade with their intended destination preserved, so signing up returns them to where they were.

Personalization for signed-in users. Setting a role at /welcome tunes the AI Analyst’s framing, the getting-started guide shown, and which content is surfaced first. A clinician sees clinical-pillar entry points; a CFO sees economics and operations. The role is stored on the account and can be changed at any time.

17. Accounts, Roles, and Subscriptions

The platform has a tiered access model enforced at the edge by the proxy (see §28). The role hierarchy, lowest to highest:

free → subscriber → student → professional → advisory → admin

What each tier sees (typical): - **Free / public** — pillar hubs, the book browser, The Wire, marketing/pricing pages, the AI Analyst widget (limited). - **Subscriber and above** — the AI Analyst full chat, the dashboards, the HTI dashboard, the advisory hub. - **Admin** — the /admin console: users, access codes, analytics, revenue, role-change log, content ingestion.

Subscriptions run through **Stripe**: checkout, the customer billing portal, team checkout, and webhooks (which keep Supabase roles in sync with subscription status). Pricing is at /pricing; upgrades at /upgrade.

Account management lives under /account: profile, billing, subscription, bookmarks, courses, developer API keys (with create/rotate/revoke), and referrals.

18. Search, Bookmarks, and Personalization

- **Search** (/search, /api/search) — across platform content.
 - **Bookmarks / My Library** (/saved, /account/bookmarks) — save any page or tool via the bookmark icon; build a personal research center.
 - **Role personalization** — setting your role at /welcome personalizes what the AI Analyst and getting-started flow surface.
 - **Chapter notes** — personal annotations on book chapters, stored to your account.
-

PART IV — THE ACADEMY

The Wire in depth. The Wire (/the-wire) is a continuously updated intelligence feed assembled from external RSS sources across all six pillars. Headlines are de-duplicated and HTML-decoded; each item is categorized by pillar so a reader can filter to, say, only Technology or only Policy news. Registered users can comment on items (threaded, with comment counts), making The Wire both a news surface and a light community layer. Operationally, the feed is cached (with a cache-bypass option for forced refresh) and surfaced via /api/wire.

Dashboards in depth. The 50-State Dashboard is the platform's comparative-evidence engine. At the national level it ranks states; drilling into /dashboard/[state] shows that state's transformation profile; drilling further into /dashboard/[state]/hospitals/[hospital] shows hospital-level financial and performance data. Vermont has a dedicated drill-down (/dashboard/vermont/hospitals) because it is the teaching case. The HTI Dashboard expresses the same data as a single composite index for quick cross-state comparison.

19. Structure: Courses, Tracks, Lessons

The Academy (/academy) is HTR's executive-education layer, built on a three-level hierarchy:

Courses → Tracks → Lessons

- **~15 courses** spanning all six pillars. Examples: Healthcare Interoperability & Data Exchange; AI & Machine Learning in Healthcare; Value-Based Care Fundamentals; Welcome & the HTR Framework.
- **~90 tracks** — a course's sub-modules (e.g., the Interoperability course contains tracks like *Foundations of Interoperability*, *FHIR APIs & Patient Access*, *EHR Systems & Integration*, and *Data Standards Beyond FHIR*).
- **~243 lessons** — the actual content units within tracks.

Where the data lives (important for maintenance): - Course/track/lesson **structure and membership** live in **Supabase** (courses, tracks, lessons tables; each row carries a pillar). - Rich lesson **body content** lives in **Sanity** as academyModule documents. - The link is lessons.sanity_slug = academyModule._id. If a lesson's sanity_slug resolves to a real Sanity doc, the full rich body renders; if it is null or points to a missing doc, the app falls back to lighter inline content blocks.

20. Using the Academy

 **For a learner:** - /academy — the landing page: the course catalog and the executive-education framing. - /academy/courses and /academy/courses/[slug] — browse courses and open the course player. - /academy/tracks/[courseSlug]/[lessonSlug] — the lesson reader. - /academy/case-studies, /academy/webinars — supplementary content. - /academy/glossary and /academy/medicaid/glossary — terminology references. - /academy/faculty — the faculty/authors.

Many lessons include “**Apply It in the [Tool]**” callouts that deep-link to the matching Research Lab tool and mode — turning a concept into hands-on modeling. For example, lessons in the *EHR Systems & Integration* track link to the EMR/EHR Lab's vendor, workflow, and data-quality modes, and a dedicated EHR business-case lesson links to its cost mode.

A typical learner journey. A new learner lands on `/academy`, picks a course aligned to their role (e.g., a CMIO chooses *Healthcare Interoperability & Data Exchange*), and works through its tracks in order. Within a track, lessons build on one another; many lessons end with an “Apply It” callout that opens the matching Research Lab tool so the learner immediately practices the concept. Progress is saved to the account and visible at `/account/courses`. On finishing a course, the learner receives a certificate (§22).

Case studies and webinars complement the structured courses: case studies (`/academy/case-studies`) work through real transformation scenarios, and webinars (`/academy/webinars`) provide recorded deep-dives. The glossary (`/academy/glossary`, plus a Medicaid-specific glossary) is a standing reference a learner can consult from any lesson.

21. Getting Started and Learning Paths

- **`/academy/getting-started`** — the onboarding hub. The recommended first session: (1) set your role at `/welcome`; (2) ask the AI Analyst your real question; (3) read your role guide; (4) explore the Research Lab; (5) bookmark your key tools. Includes role-specific quick-starts for executives, policy analysts, clinicians, economists, health-tech professionals, compliance officers, researchers, and investors.
- **`/academy/getting-started/research-lab`** — the canonical “how to use each tool” guide: who each tool is for, when to use it, and one power tip per tool. This is the user-facing companion to §14 of this manual.
- **`/academy/personalized-learning`** — a tailored learning path that sequences lessons to the learner’s role and goals.

22. Certificates and Verification

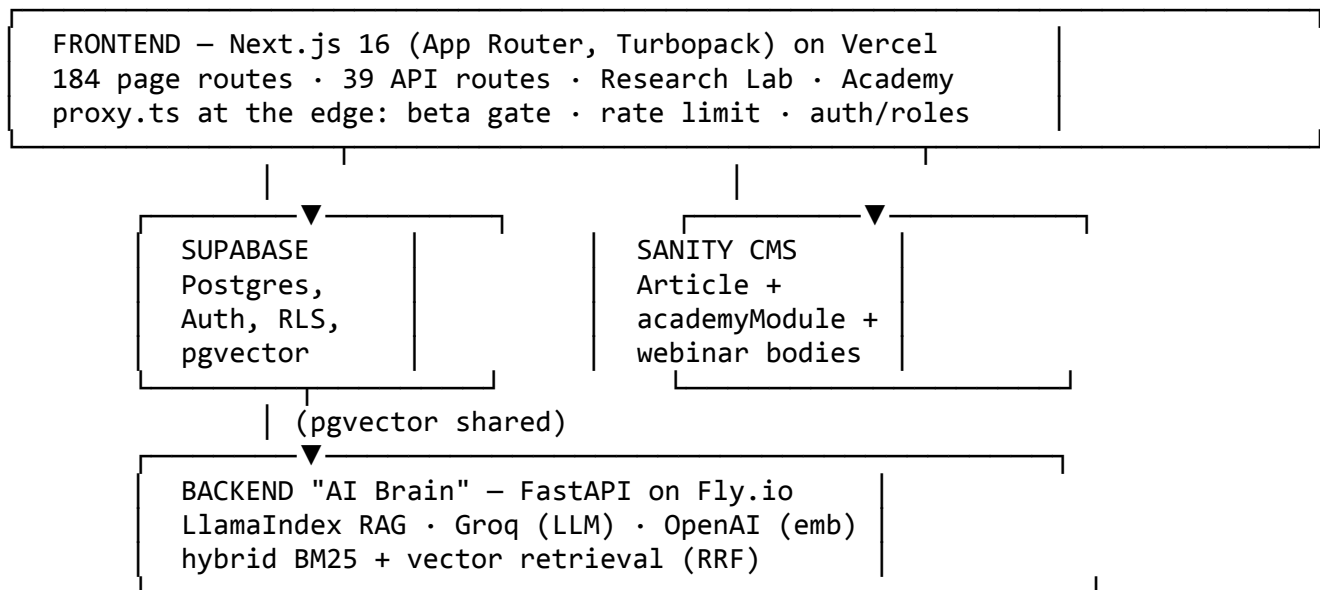
The Academy tracks course completion and issues certificates (`/api/academy/certificates`). Completed certificates are publicly verifiable at `/verify/[hash]` (the endpoint is rate-limited at 20 requests/IP/minute to deter scraping). A learner’s course progress is visible under `/account/courses`.

PART V — TECHNICAL ARCHITECTURE

A note on personalization and learning paths under the hood. The personalized-learning feature (`/academy/personalized-learning`, backed by `/api/personalized-learning`) assembles a recommended lesson sequence for a learner based on their role and goals, and can validate the platform links it generates against the live Sanity catalog so it never points a learner at a missing module. An audio companion (`/api/personalized-learning/audio`) can narrate the path. This is the most “AI-adjacent” Academy feature and shares the backend’s content awareness.

23. System Overview

HTR is a three-tier system spanning two hosts and two managed data services:



The frontend is the user-facing application and the system of record for routing and access control. The backend is a dedicated AI service. Supabase holds structured/relational data and the vector store; Sanity holds long-form editorial content.

24. The Frontend

- **Framework:** Next.js 16.2 (App Router, Turbopack), React 19, TypeScript.
- **Styling:** Tailwind CSS with styled-components.
- **Error monitoring:** Sentry (`@sentry/nextjs`); server and React Server Component errors are captured via `instrumentation.ts`.
- **Key locations:**
 - `frontend/app/` — routes. Page routes are `page.tsx`; API routes are `route.ts` under `app/api/`.
 - `frontend/components/` — shared components. `components/research/` holds the 24 Research Lab tool components.
 - `frontend/lib/taxonomy/` — the canonical pillars/tools/chapters spine (see §27).
 - `frontend/lib/data/` — pillar topics, learning-track data.
 - `frontend/proxy.ts` — the edge proxy (auth, roles, beta gate).

-  frontend/sanity/ — the embedded Sanity Studio (served at /studio) and the schema types.
-  frontend/next.config.ts — config, security headers (CSP, HSTS), image remote patterns, outputFileTracingExcludes, and turbopack.root.
- **Build/verify scripts:** npm run typecheck, npm run lint, npm run build, npm run bundle:check, and npm run smoke (all of the above together).

25. The Backend (AI Brain)

- **Framework:** FastAPI (Python), served by uvicorn.
- **RAG stack:** LlamaIndex core; Groq for the LLM, OpenAI for embeddings; hybrid retrieval (BM25 + vector with reciprocal-rank fusion); sentence-window chunking (window size 3).
- **Vector store:** pgvector inside the Supabase Postgres instance.
- **Knowledge sources:**
 -  backend/data/ — the book, Vermont source documents (Act 167/68, the Wyman Report, the RHT application/budget, 50-state spotlights), and HTR_ResearchLab_Documentation.pdf.
 - Sanity CMS content (articles, academy modules), ingested via GROQ.
 -  backend/platform_catalog.py — the canonical catalog of every page, tool, and simulator, used to build the AI's tool awareness and the platform card grid.
- **Entry point:**  backend/main.py. Health: GET /health reports index_ready, the active model, embedding model, vector store, and retrieval mode.
- **Indexing:**  backend/services/indexing.py ingests PDFs + Sanity at startup and on demand via the ingest endpoint.  backend/services/catalog_search.py embeds the catalog so the Analyst can semantically match a question to a tool.
- **The frontend /api/chat route is a thin proxy** to POST /api/chat on this backend; auth/tier is validated backend-side.

26. Data Stores: Supabase and Sanity

Supabase (Postgres + Auth + pgvector): - **Auth & access:** user accounts, user_roles, subscription state. - **Academy:** courses, tracks, lessons. - **Operational tables:** bookmarks, chapter notes, wire comments, ticker cache, HTI scores, and more. - **Migrations:**  supabase/migrations/ — **36 SQL migrations, append-only.** ⚠️ Never edit a shipped migration file; always add a new, higher-numbered one. Editing shipped migrations corrupts the migration history for any environment that already ran them. - **Security:** Row-Level Security protects user-scoped data. The SUPABASE_SERVICE_ROLE_KEY bypasses RLS and is **server-only** — never expose it to the client.

Sanity CMS: - **Content types:** 22 schema types in  frontend/sanity/schemaTypes/, including policyAnalysis (articles), academyModule (lesson bodies), webinar, and case studies. - **Access:** queried via GROQ; reads are public for published content; writes use SANITY_API_TOKEN (server-only). - **The Academy link:** lessons.sanity_slug (Supabase) = academyModule._id (Sanity). - **Studio:** the editing UI is embedded in the app at /studio.

27. The Taxonomy Spine

 frontend/lib/taxonomy/ is the **single source of truth** that keeps the three components aligned:

- **pillars.ts** — the six pillars.
- **tools.ts** — the canonical tool registry: each entry is { id, label, href, pillars[], chapters?[], desc? }. Every tool surface (sidebar, hub, book chapter links, AI catalog cross-references) resolves a tool by id via `getTool(id)`. This file is also what the book's Appendix E is generated/checked against.
- **chapters.ts** — the book's chapters and each chapter's `platformLinks` (the tool IDs that appear in that chapter's "Work This Chapter" table).

The discipline is simple but strict: **a tool is only "real" when it appears consistently in the registry, every navigation surface, the AI catalog, the book chapter tables, and the Academy.** §34 is the checklist that enforces this.

28. Authentication, Roles, and the Proxy

 `frontend/proxy.ts` (Next 16's renamed "middleware" convention) runs at the edge on every request and performs three checks in order:

1. **Beta gate.** Any visitor without the `htr_beta` cookie is redirected to `/beta` (an access-code wall). Exemptions: `/beta`, `/api/*`, `/studio`, and static assets.  **This is currently unconditional** — there is no env switch to disable it. For a public launch it must be removed or made toggleable (see §38).
2. **Rate limiting.** The public `/verify/*` certificate endpoint is limited to 20 requests/IP/minute.
3. **Auth + role enforcement.** For protected prefixes — `/admin`, `/dashboard`, `/chat`, `/hti-dashboard`, `/advisory-hub`, `/account`, `/onboarding` — it reads the Supabase session and compares the user's role against the required tier in the hierarchy `free` → `subscriber` → `student` → `professional` → `advisory` → `admin`. To avoid a DB lookup on every request, the resolved role is cached in an HMAC-signed cookie (`MIDDLEWARE_ROLE_SECRET`) for one hour. The `ALLOW_AUTH_BYPASS=true` env flag (default off) opens role-gated routes during beta and must be unset for GA.

Logged-in users hitting `/login` or `/signup` are redirected to `/account`; unauthenticated users hitting a protected route are sent to `/login?from=...`; under-privileged users are sent to `/upgrade?from=...`

29. Request Flows (Worked Examples)

A public visitor opens a pillar hub (`/technology`): proxy checks the beta cookie (must be present post-launch) → route is not protected → page renders server-side, pulling pillar content from `pillar-topics.ts` and any editorial from Sanity.

A subscriber asks the AI Analyst a question: the chat UI calls `/api/chat` → the proxy confirms the subscriber role → the route proxies to the FastAPI backend → the backend runs hybrid retrieval over the corpus + catalog, composes an answer with the LLM, and streams it back → the UI renders the answer plus any tool/page links.

A learner opens a lesson: route loads the Supabase lessons row → reads `sanity_slug` → fetches the `academyModule` body from Sanity → renders the rich body (or falls back to inline blocks if the slug doesn't resolve).

A user subscribes: Stripe checkout → Stripe webhook hits `/api/stripe/webhook` → the handler updates the user's role in Supabase → the next request's proxy role-check (cache miss) reads the new role and grants access.

PART VI — ENVIRONMENT & CONFIGURATION

30. Environment Variables (Complete Reference)

⚠️ Secrets are never committed to git. Set them in the host: Vercel for the frontend, Fly for the backend. Any `NEXT_PUBLIC_*` variable is exposed to the browser — never put a secret value in one.

Supabase - `NEXT_PUBLIC_SUPABASE_URL` (*public*) — project URL. -
`NEXT_PUBLIC_SUPABASE_ANON_KEY` (*public*) — anon key; safe client-side, RLS-protected. -
`SUPABASE_SERVICE_ROLE_KEY` (***secret, server-only***) — full DB access; bypasses RLS.

Sanity - `NEXT_PUBLIC_SANITY_PROJECT_ID`, `NEXT_PUBLIC_SANITY_DATASET` (*public*). -
`SANITY_API_TOKEN` (***secret***) — content writes/queries.

AI backend - `NEXT_PUBLIC_BACKEND_URL` / `BACKEND_URL` / `PYTHON_BACKEND_URL` — the FastAPI backend address (the frontend `/api/chat` proxy target). - (*Backend-side*) Groq + OpenAI API keys; the Postgres/pgvector connection string.

Auth & security - `MIDDLEWARE_ROLE_SECRET` (***secret***) — signs the role-cache cookie. -
`ALLOW_AUTH_BYPASS` — "true" opens role-gated routes; **beta only**, unset for GA. -
`API_KEY_HMAC_SECRET` (***secret***) — signs developer API keys.

Stripe (billing) - `STRIPE_SECRET_KEY`, `STRIPE_WEBHOOK_SECRET` (***secrets***). - `STRIPE_PRICE_*` — price IDs for each tier × interval: SUBSCRIBER, STUDENT, PROFESSIONAL, TEAM, each `_MONTHLY` and `_YEARLY`.

Email & lifecycle - `LOOPS_API_KEY` and `LOOPS_TEMPLATE_*` (welcome, digest, payment-failed, survey-results, trial-ending). - `RESEND_API_KEY`.

Jobs & feeds - `CRON_SECRET`, `DIGEST_SECRET`, `INGEST_SECRET` — shared secrets guarding the cron, digest, and ingest endpoints. - `TICKER_API_URL`.

Site - `NEXT_PUBLIC_SITE_URL` / `NEXT_PUBLIC_APP_URL` — canonical URLs (redirects, emails; should back metadataBase). - `NODE_ENV`.

31. Deployment Topology and Constraints

Frontend → Vercel. Build runs `cd frontend && next build`. Config in  `frontend/vercel.json` and the repo-root `vercel.json`.

⚠️ **Two hard-won constraints encoded in `next.config.ts` — do not remove:** 1. `outputFileTracingExcludes` keeps `*.m4a`, `*.mp3`, and `*.pdf` out of serverless function bundles. The `/read/[slug]` route reads narration `.txt` files from `public/` at runtime; without this exclusion Next's file tracer pulls all ~280 MB of narration audio into that function, exceeding Vercel's **300 MB function limit** and **failing the deploy**. With it, the function is ~24 MB. 2. `turbopack.root` is pinned to the frontend directory because stray `package-lock.json` files exist higher in the tree (home dir, repo root); without the pin, Next infers the wrong workspace root and warns.

Backend → Fly.io (authoritative;  `backend/fly.toml`).  `backend/Procfile` and `backend/railway.toml` exist as alternative/fallback descriptors. Full instructions in  `backend/DEPLOYMENT.md`.

Databases → **Supabase** (managed Postgres + auth + pgvector) and **Sanity** (managed CMS) — both SaaS, no self-hosting.

PART VII — OPERATIONS & MAINTENANCE

32. Routine Maintenance Cadence

Task	Cadence	How
Dependency security review	Monthly	<code>npm audit</code> (frontend), <code>pip-audit</code> (backend). Patch high/critical with direct exposure; transitive build-tooling vulns can be deferred with a note.
Supabase backup	≤ 30 days	Managed PITR + a manual export.
Sanity dataset export	≤ 30 days	Store under <code>sanity-backups/</code> .
Sentry triage	Weekly	Review open clusters; fix the top recurring errors.
Typecheck + lint	Every change	<code>npm run typecheck</code> , <code>npm run lint</code> .
Bundle budget	Every release	<code>npm run bundle:check</code> (budget 20 MB; largest chunk < 1.5 MB).
Full smoke	Every release	<code>npm run smoke</code> .
Backend index freshness	After any corpus/catalog change	Restart backend or trigger ingest (see §35).

33. Content Operations

Editorial / Academy content lives in Sanity (bodies) and Supabase (structure). The standing rules:

-  **Academy seed scripts are upsert-only — they never delete.** To remove a lesson, delete the Supabase row directly, then check all tracks for orphans/dupes.
-  After posting a lesson body to Sanity, **set the Supabase lesson's `sanity_slug`** (= the `Sanity_id`) or the app renders thin fallback content instead of the rich body.
-  Content-write scripts must live in `frontend/scripts/` (so Node resolves `@supabase/supabase-js`), not `/tmp`.
-  **Broken/missing `sanity_slug` links and orphaned modules are sometimes intentional** — content deliberately disconnected because it was unverified and pulled. An audit that flags such a gap is raising a question, not issuing a work order. Never auto-restore or re-link without explicit sign-off and proof the content is verifiable. Editorial content is the owner's domain.

Editing the book — edit BOTH `HTR_Book_v41.docx` (carefully; formatting-sensitive) and `HTR_Book_v41.md`, kept in sync; regenerate the `.pdf` when a distributable is needed. Distinguish the **book** (manuscript) from the **/book page** (frontend code).

34. Adding a Research Lab Tool (Full Checklist)

This is the most cross-cutting operation in the platform; missing a step leaves the tool half-wired. Do all of it:

1.  `frontend/lib/taxonomy/tools.ts` — add the registry entry (source of truth).
2.  `frontend/app/research-lab/<bench>/<Bench>Client.tsx` — add the tab: a TABS entry, a `dynamic()` import, and the render line.
3.  `frontend/app/research-lab/ResearchLabHub.tsx` — add the hub card to the bench's section AND the renderer case.
4.  `frontend/components/research/<Tool>.tsx` — build the component (client-side; match the existing tool pattern, including an “illustrative model” disclaimer).
5.  `frontend/components/HomeSidebar.tsx` — add the tool ID to the right pillar's `labToolIds`. **Easily missed** — without it the tool is absent from sidebar nav.
6.  `backend/platform_catalog.py` — add a catalog entry with rich keywords. **Easily missed** — this is what lets the AI Analyst surface and recommend the tool.
7.  `frontend/lib/taxonomy/chapters.ts` — add the tool ID to the relevant chapter's `platformLinks`.
8.  `frontend/lib/data/pillar-topics.ts` — mention it in the pillar's tool footer.
9.  `frontend/app/academy/getting-started/research-lab/page.tsx` — add a who/when/tip guide entry.
10. **The book manuscript** — add the tool to the chapter's “Work This Chapter” table AND Appendix E, in BOTH `.docx` and `.md`.
11. **Tool count** — update the count copy everywhere (currently 24): the hub, getting-started, pillar footers, header mega-menu, home card, advisory page, the AI corpus doc, and Appendix E.

 **The two most-forgotten surfaces are #5 (sidebar) and #6 (AI catalog).** A tool can pass typecheck and render on its bench while being invisible to navigation and unknown to the Analyst.

After wiring: `npm run typecheck`, lint the new files, confirm the AI catalog parses and matches a sample query, and load the route to confirm it renders.

35. The AI Knowledge Corpus

The AI Analyst's knowledge has two maintainable parts:

1. **Tool/page catalog** —  `backend/platform_catalog.py`. A Python list; edit directly. Used for tool recommendation and the platform card grid. Updating this is part of §34.
2. **Document corpus** — PDFs in  `backend/data/` (book, Vermont docs, `HTR_ResearchLab_Documentation.pdf`) plus Sanity content. When a source changes, regenerate its PDF (e.g., via pandoc from the `.md` source) and **re-ingest**.

⚠ The backend builds its semantic index **at startup**. Editing a catalog or corpus file does not change retrieval until the index rebuilds — restart the backend or trigger the ingest endpoint. Verify with `GET /health (index_ready: true)`.

Worked example — keeping the Analyst current after adding tools: edit

`platform_catalog.py` (catalog) and

`backend/data/HTR_ResearchLab_Documentation.md` → regenerate its `.pdf` →

restart/re-ingest the backend → confirm a query like “compare EHR vendors” returns the EMR/EHR Lab.

36. Monitoring and Incident Response

- **Errors:** Sentry for both tiers; server/RSC errors captured via `instrumentation.ts`.
- **Health checks:** backend `GET /health` (reports `index_ready` and config); frontend `GET /api/health`; in-app `/system-vitals`.
- **Billing:** monitor Stripe webhook deliveries; a failed `/api/stripe/webhook` means roles can drift from subscription status.
- **Common incidents:**
 - *AI Analyst not responding* → check the backend is up (`/health`), `BACKEND_URL` is correct, and the LLM/embedding keys are valid. `/api/chat` returns a clear error if it can't reach the backend.
 - *Users locked out unexpectedly* → check the beta gate and `ALLOW_AUTH_BYPASS` state, and that the role-cache secret is set.
 - *Deploy fails on function size* → something re-introduced large files into a function bundle; verify `outputFileTracingExcludes` (§31).
 - *Lessons show thin content* → a `sanity_slug` is missing or broken (§33).

37. Backups and Disaster Recovery

- **Supabase:** managed point-in-time backups plus periodic manual exports; schema reconstructible from `supabase/migrations/`.
- **Sanity:** periodic dataset exports under `sanity-backups/`.
- **Code:** GitHub `bbensaid/VHP`, main branch.
- **Book .docx:** gitignored — back up separately (round-trips through Google Docs).
- **Recovery order after a total loss:** restore code (GitHub) → restore Supabase (backup + migrations) → restore Sanity (dataset export) → redeploy frontend (Vercel) and backend (Fly) → rebuild the AI index → smoke-test.

PART VIII — LAUNCH

Why this matters more in a paused-development posture. Once active development stops, the platform becomes a static target: no one is watching the build daily, and the most likely sources of breakage are (a) external dependency CVEs, (b) expiring credentials/API keys (Stripe, Sanity token, LLM keys), and (c) drift between the running backend index and the content. The maintenance cadence in §32 is therefore not optional even in maintenance mode — it is the difference between a system that ages gracefully and one that silently degrades. Assign an owner for the monthly security review and the quarterly credential check before stepping away.

38. Pre-Launch Checklist

The one true blocker is the beta gate. Everything in  must be done before going public.

 **Required** - [] **Disable the beta gate.** As shipped, `proxy.ts` redirects every visitor to `/beta`; the public cannot enter without an access code, and there is **no env switch** for it. Action: add a production toggle (e.g., `NEXT_PUBLIC_PUBLIC_LAUNCH=true`) that bypasses the gate, or remove the gate block. (*Code change required.*) - [] Confirm `ALLOW_AUTH_BYPASS` is **unset** in production (so role-gated routes are actually protected). - [] All required env vars set in Vercel and Fly (§30) — especially Stripe **live** keys and price IDs if billing is live at launch. - [] A green production deploy (the 300 MB function fix and proxy migration are in; confirm the latest build deploys clean). - [] `MIDDLEWARE_ROLE_SECRET` set (role caching + integrity).

 **Strongly recommended (launch quality)** - [] Add `metadataBase` + Open Graph / Twitter card metadata (with a share image) in `app/layout.tsx`, so shared book/platform links render proper previews — important for a public book launch. - [] Add `app/robots.ts` (pointing at the existing `sitemap.ts`) and a favicon / `app/icon`. - [] `npm audit` — patch any directly-exposed high/critical vulns (the current 4 highs are transitive/build-tooling: `vite`, `undici`, `ws`, `form-data`). - [] Rebuild the backend index so the Analyst knows all 24 tools (catalog + corpus current). - [] Logged-out spot check: `/`, `/book`, `/research-lab`, each pillar hub, `/academy`, `/the-wire` all load.

 **Verify-only (already true)** - [x] No secrets committed to git; no secrets in client components. - [x] Book, platform, and Academy aligned (24 tools consistent across registry, sidebar, hub, AI catalog, chapter tables, Appendix E). - [x] Tool counts unified to 24.

39. Go-Live Runbook

1. Freeze content; take fresh Supabase + Sanity backups.
2. Set/confirm production env vars; flip the beta-gate toggle off; confirm `ALLOW_AUTH_BYPASS` unset.
3. Deploy frontend (Vercel) and backend (Fly); confirm both healthy (`/api/health`, `/health` with `index_ready: true`).
4. Rebuild/confirm the AI index.
5. Smoke-test public routes logged out, then one full subscriber journey: sign up → pay (Stripe) → access a gated tool/chat.
6. Confirm Stripe webhooks deliver and update roles.
7. Announce.

40. Post-Launch Operations

- Watch Sentry closely for the first 24–48 hours.

- Monitor Stripe (new subscriptions, failed payments) and the lifecycle emails (welcome, trial-ending).
- Watch backend latency and `index_ready`.
- Resume the §32 maintenance cadence.
- Since active development is paused: keep the security cadence (monthly `npm audit/pip-audit`) even in maintenance mode — dependency CVEs are the main risk to a static deployment.

APPENDICES

Appendix A — Complete Tool Catalog

Research Lab — 24 hub tools (by bench):

Interoperability & Risk: FHIR Interoperability Lab; Risk Stratification Engine; EMR/EHR Lab; Statewide EHR Deployment Modeler. *Payment Models & VBC:* APM Design Lab; Shared Savings Calculator; CEA Calculator; Global Budget Transition Modeler. *Policy & Quality Sciences:* Policy Simulator; Clinical Quality Optimizer; Hospital Financial Stress Test; HTA Studio; Actuarial Lab; Work Requirements Calculator; H.R. 1 Cliff Scenario. *Population & Equity:* Population Health Modeler; Health Equity Studio. *Technology & AI:* AI Clinical Governance Lab; Digital Health Lab. *VBC, Clinical & Quality:* Clinical Data Exchange Lab; Risk Stratification Methodology; VBC Quality Measures; High vs. Low Value Care. *Knowledge & Workspace:* Transformation Scorecard; VBC Readiness Assessment; Evidence Library; Workforce Modeler; Innovation Leaderboard; Research Workspace; CIN & Shared Services Modeler; EMS Transformation Modeler.

Top-level simulators & dashboards (registry, not hub cards): HTR Simulator; Transformation Friction Index; Impact Simulation; HTI Dashboard; Investment Tracker; Medicaid Eligibility Simulator; Six-Pillar Map; The Wire.

The “24” headline is the count of tool cards in the Research Lab hub. The full `tools.ts` registry total is higher because it also lists the top-level simulators and dashboards above.

Appendix B — Route Map

Area	Routes
Home / brand	/
Book	/book, /read/[slug], /book/listen
Pillars	/policy, /economics, /technology, /clinical, /equity, /operations (+ /[pillar]/[slug] sub-pages)
Research Lab	/research-lab + /research-lab/{interoperability, payment-models,policy-quality, population-equity,technology-ai, vbc-clinical-quality,knowledge -workspace}

Area	Routes
AI Analyst	/chat (widget everywhere)
Academy	/academy, /academy/courses, /academy/courses/[slug], /academy/tracks/[courseSlug]/[lessonSlug], /academy/getting-started, /academy/getting-started/research-lab, /academy/case-studies, /academy/webinars, /academy/glossary
Dashboards / sims	/dashboard, /dashboard/[state], /dashboard/[state]/hospitals/[hospital], /hti-dashboard, /htr-simulator, /transformation-friction-index, /impact-simulation, /investment-tracker, /medicaid-eligibility-simulator
The Wire	/the-wire
Vermont & states	/vermont-rht-program, /vermont-act-167, /vermont-act-68, /ahead-model, /vermont-blueprint, /vermont-vcci, /vermont-sash, /vermont-sdoh, /oregon-cco, /california-calaim, /states/[state], /compare-states
Account / billing	/account/*, /login, /signup, /pricing, /upgrade, /welcome
Admin	/admin, /admin/{users,access-codes,analytics,revenue,role-changes,ingest}
CMS	/studio
Key APIs	/api/chat, /api/search, /api/stripe/{checkout,portal,webhook}, /api/academy/certificates, /api/wire, /api/health

Appendix C — Environment Variable Quick Table

Variable	Public?	Purpose
NEXT_PUBLIC_SUPABASE_URL / _ANON_KEY	Yes	Supabase client

Variable	Public?	Purpose
SUPABASE_SERVICE_ROLE_KEY	Secret	Server DB access
NEXT_PUBLIC_SANITY_PROJECT_ID / _DATASET	Yes	Sanity client
SANITY_API_TOKEN	Secret	Sanity writes
BACKEND_URL / NEXT_PUBLIC_BACKEND_URL	Mixed	AI backend address
MIDDLEWARE_ROLE_SECRET	Secret	Role-cookie signing
ALLOW_AUTH_BYPASS	Config	Beta route bypass (unset for GA)
API_KEY_HMAC_SECRET	Secret	Developer API keys
STRIPE_SECRET_KEY / STRIPE_WEBHOOK_SECRET	Secret	Billing
STRIPE_PRICE_*	Config	Tier price IDs
LOOPS_API_KEY / RESEND_API_KEY	Secret	Email
CRON_SECRET / DIGEST_SECRET / INGEST_SECRET	Secret	Job endpoints
NEXT_PUBLIC_SITE_URL / _APP_URL	Yes	Canonical URLs

Appendix D — Glossary

- **AHEAD Model** — CMS all-payer global-budget model; Vermont’s Medicare partner to Act 68 (performance year from Jan 2028).
- **Act 167 (2022) / Act 68 (2025)** — Vermont’s enabling, then mandatory, transformation legislation.
- **APM** — Alternative Payment Model.
- **CIN** — Clinically Integrated Network.
- **CMMI** — CMS Center for Medicare & Medicaid Innovation.
- **FHIR** — Fast Healthcare Interoperability Resources (HL7 standard).
- **HCC / RAF** — Hierarchical Condition Categories / Risk Adjustment Factor.
- **HTI** — Health Transformation Index.
- **RAG** — Retrieval-Augmented Generation (the AI Analyst’s method).
- **RHT** — Rural Health Transformation Program (\$195M federal award to Vermont).
- **RLS** — Row-Level Security (Supabase).
- **RRF** — Reciprocal Rank Fusion (hybrid-retrieval ranking).
- **TCOC** — Total Cost of Care.
- **USCDI** — United States Core Data for Interoperability.
- **VHCURES** — Vermont’s all-payer claims database.
- **VITL / VHIE** — Vermont’s health information exchange.

Appendix E — Maintenance Quick Reference

If you...	Do this	Section
Add a Research Lab tool	Follow the 11-step checklist	§34
Edit the book	Change BOTH .docx and .md; regenerate .pdf	§6, §33
Change AI knowledge	Edit platform_catalog.py and/or backend/data/PDFs; re-ingest	§35
Change the DB schema	Add a NEW migration (never edit shipped ones)	§26
Remove an Academy lesson	Delete the Supabase row directly (seeds are upsert-only)	§33
Deploy	Vercel (frontend) / Fly (backend); keep outputFileTracingExcludes	§31
Go public	Disable the beta gate; run the pre-launch checklist	§38–39

End of guide. This document should be updated alongside the platform — in particular the tool count, Appendix A, and §34 — whenever the Research Lab changes. Maintained by the HTR team.

PART IX — WORKED EXAMPLES & EXTENDED REFERENCE

This part adds hands-on, numeric walkthroughs for the most-used tools, end-to-end user scenarios by role, expanded troubleshooting playbooks, and a fuller data-model reference. It is intended for readers who want to see exactly what a tool does with concrete inputs, and for operators who need deeper detail than Parts VII–VIII.

41. Worked Tool Examples

Each example gives representative inputs and the kind of output you should expect. (All figures are illustrative — the tools are planning models, not live data.)

41.1 EMR/EHR Lab — building an EHR business case

Scenario: a 14-hospital system evaluating whether to standardize on a single vendor.

 **Step 1 — Vendor Comparison.** Open `/research-lab/interoperability?tab=emr`. Set the criteria weights to reflect your priorities — e.g., Interoperability 35, Usability 25, Cost 25, Ambulatory fit 15. Observe how the four vendors re-rank: a high interoperability weight favors Epic; a high cost weight favors MEDITECH/athenahealth. Select the vendor you want to carry forward (say, Oracle Health).

 **Step 2 — Adoption & Cost.** Switch to the Cost mode (or open `&mode=cost`). Enter: 14 hospitals, 1,800 providers, a 24-month rollout, \$550K average clinical revenue/provider. The tool returns an implementation capex, an annual license/support figure, a modeled go-live productivity dip (a temporary revenue loss as clinicians ramp), and a 5-year cost-vs-benefit table with a **break-even year**. The headline question it answers: *in which year does cumulative benefit overtake cumulative cost?*

 **Step 3 — Data Quality.** Switch to the Data Quality mode. The mock record is seeded by the selected vendor's interoperability score; review which of the ten USCDI classes are complete vs. gapped. Higher-interoperability vendors show fewer gaps.

 **Step 4 — Workflow Simulator.** Set a primary-care panel (e.g., 20 patients/day) and toggle interventions — ambient AI documentation, scribes, template redesign. Watch projected daily EHR minutes move relative to the ~360 min/day national benchmark. This quantifies the “pajama time” burden and the ROI of documentation-burden interventions.

What you walk away with: a defensible cost/benefit narrative, a data-quality expectation, and a clinician-burden estimate — the three legs of an EHR decision.

41.2 Statewide EHR Deployment Modeler — the Act 167 question

Scenario: answering whether Vermont should pursue a single statewide EHR or invest in FHIR interoperability.

 Open `/research-lab/interoperability?tab=statewide-ehr`. Enter the number of participating hospitals and the per-hospital migration vs. integration cost assumptions. The tool produces two paths side by side — *Single statewide EHR* vs. *FHIR interoperability* — each scored on 10-year total cost, data timeliness, migration disruption, and vendor lock-in, and

renders a verdict. The pedagogical point (straight from Chapter 4): if FHIR delivers acceptable real-time exchange, the case for a disruptive single-EHR migration weakens.

41.3 CIN & Shared Services Modeler — the administrative cost gap

Scenario: quantifying the savings from the RHT-funded Clinically Integrated Network.

 Open </research-lab/knowledge-workspace?tab=cin>. Enter 14 hospitals, total annual discharges, and the administrative cost premium (anchored at ~\$1,303/discharge). Toggle which functions to consolidate — billing, coding, credentialing, HR, IT — and set combined group-purchasing volume. The tool returns annual admin savings, group-purchasing savings, the CIN's setup and operating cost, and a **break-even in months**. The lesson: shared services only pay off at sufficient scale; the tool shows where that threshold is.

41.4 EMS Transformation Modeler — fragmentation to margin

 Open </research-lab/knowledge-workspace?tab=ems>. Enter Vermont's 31 EMS agencies, a target number of regions, annual 911 volume, the non-emergent share, and community-paramedicine reach. The tool returns ED visits diverted, admissions prevented, regionalization savings, the program cost, and net annual impact — illustrating how treat-and-refer converts fragmentation into margin under global budgets.

41.5 HTR Simulator — the dependency cascade

 Open </htr-simulator>. Build a profile with strong Economics ambition but weak Technology and Policy scores. The composite readiness collapses — not because the Economics inputs are bad, but because the simulator enforces the dependency logic. Then raise Technology and Policy first and watch the same Economics inputs finally produce a viable score. *That reversal is the central argument of the book, made tangible.*

41.6 Hospital Financial Stress Test — surviving the transition

 Open </research-lab/policy-quality?tab=scorecard>. Select a peer archetype (CAH, Rural PPS, Urban Tertiary), then apply stressors: a payer-mix shift toward Medicaid, a Medicaid rate cut, and a volume decline. The 10-year projection shows the margin trajectory — reproducing the Oliver Wyman finding that most Vermont hospitals reach operating loss under conservative assumptions absent transformation.

42. End-to-End Scenarios by Role

A hospital CFO preparing for AHEAD (Jan 2028): 1. Read Chapter 6–7 (Economics) via </book>. 2. Run the Hospital Financial Stress Test to see the baseline trajectory. 3. Run the Shared Savings Calculator and Global Budget Transition Modeler to model the AHEAD environment. 4. Run the CIN & Shared Services Modeler to size administrative savings. 5. Take the VBC Readiness Assessment; export the gap analysis. 6. Ask the AI Analyst to summarize the combined picture.

A state policy analyst designing a waiver: 1. Read Chapters 2–3 (Policy). 2. Run the Policy Simulator against a pre-loaded state scenario. 3. Run the Work Requirements Calculator and H.R. 1 Cliff Scenario for coverage impact. 4. Compare against Oregon CCO and California CalAIM pages.

A CMIO evaluating EHR strategy: 1. Read Chapters 4–5 (Technology). 2. Work through the EMR/EHR Lab (§41.1) and the Statewide EHR Modeler (§41.2). 3. Take the relevant Academy track (*EHR Systems & Integration*) and the EHR business-case lesson. 4. Validate FHIR readiness in the FHIR Interoperability Lab.

43. Extended Troubleshooting Playbook

Symptom	Likely cause	Resolution
Public users hit an access-code wall	Beta gate still on	Disable the gate / set the launch toggle (§38).
Role-gated route open to everyone	ALLOW_AUTH_BYPASS=true in prod	Unset it; redeploy.
Users get logged out / role wrong	MIDDLEWARE_ROLE_SECRET missing or rotated	Set a stable secret; role cache falls back to DB if absent.
AI Analyst returns an error	Backend down / wrong BACKEND_URL / bad LLM keys	Check /health; verify env; check Groq/OpenAI keys.
AI Analyst gives stale tool info	Catalog/corpus not re-indexed	Rebuild index (§35); confirm index_ready.
New tool missing from sidebar	labToolIds not updated	Add the ID in HomeSidebar.tsx (§34 step 5).
New tool unknown to the Analyst	platform_catalog.py not updated	Add the catalog entry (§34 step 6); re-ingest.
Lesson shows thin content	sanity_slug null/broken	Set/repair the slug (§33).
Deploy fails on function size	Large files traced into a function	Verify outputFileTracingExcludes (§31).
Build warns about workspace root	Stray lockfiles	Confirm turbopack.root pin (§31).
Shared link shows no preview	metadataBase/OG missing	Add OG metadata (§38).
Stripe role not updating	Webhook failing	Check /api/stripe/webhook deliveries + STRIPE_WEBHOOK_SECRET.
Book page tool link 404s	tools.ts href / chapter link mismatch	Verify the tool id resolves via getTool (§27).

44. Extended Data-Model Reference

Supabase — principal tables (representative):

Table	Purpose	Key fields
<code>user_roles</code>	Tier per user	<code>user_id</code> , <code>role</code>
<code>courses</code>	Academy courses	<code>id</code> , <code>title</code> , <code>slug</code> , <code>pillar</code>
<code>tracks</code>	Course sub-modules	<code>id</code> , <code>title</code> , <code>slug</code> , <code>course_id</code> , <code>pillar</code>
<code>lessons</code>	Lesson units	<code>id</code> , <code>title</code> , <code>slug</code> , <code>track_id</code> , <code>pillar</code> , <code>sanity_slug</code> , <code>is_published</code> , <code>order</code> , <code>objectives</code>
<code>bookmarks</code>	Saved pages/tools	<code>user_id</code> , <code>target ref</code>
<code>chapter_notes</code>	Book annotations	<code>user_id</code> , <code>chapter ref</code> , <code>note</code>
<code>wire_comments</code>	The Wire comments	<code>thread/comment refs</code>
<code>hti_scores</code>	Health Transformation Index	<code>state/system scores</code>
<code>ticker_cache</code>	Cached ticker data	<code>cache rows</code>

Relationships: `courses 1-* tracks 1-* lessons`. Every level carries `pillar`. The cross-system join is `lessons.sanity_slug → academyModule._id` in Sanity.

Sanity — principal document types (representative of the 22): `policyAnalysis` (articles), `academyModule` (lesson bodies), `webinar`, `case-study types`, plus supporting object/block types. Bodies use a shared block-content schema (portable text) with custom blocks for images, callouts, and embeds.

The taxonomy spine (code, not DB): `pillars.ts` (6 entries), `tools.ts` (the tool registry — 24 hub tools + top-level simulators), `chapters.ts` (16 chapters + `platformLinks`). These three files are the contract that the book, platform, and Academy all honor.

45. Security Posture Summary

- **Secrets:** never in git; never in `NEXT_PUBLIC_*`; service-role and API tokens are server-only.
- **Transport:** HSTS (2-year max-age, preload), enforced in `next.config.ts` security headers.
- **CSP:** a strict Content-Security-Policy allowlists only the required origins (Supabase, Sanity, Stripe, YouTube/Wikimedia media, OSM tiles, Sentry).
- **Other headers:** `X-Frame-Options: SAMEORIGIN`, `X-Content-Type-Options: nosniff`, a restrictive `Referrer-Policy` and `Permissions-Policy`.
- **Data:** Supabase RLS protects user-scoped rows; the role cache is HMAC-signed and tamper-evident.
- **Rate limiting:** the public verify endpoint is IP-rate-limited.

- **Dependency risk:** the main residual risk in a paused-development posture is dependency CVEs — keep the monthly audit cadence.
-

End of extended reference. Combined with Parts I–VIII, this guide is the complete user, technical, and operational manual for the HTR ecosystem.

Appendix F — Frequently Asked Questions

General

What is the difference between HTR and HTS? Health Transformation Review (HTR) is the knowledge/analytical brand; Health Transformation Solutions (HTS) is the advisory-services brand. The platform supports both and resolves the displayed brand by host.

Do I need an account to use HTR? No. The book browser, pillar hubs, The Wire, and a limited AI Analyst are public. An account (and a subscription for higher tiers) unlocks the full AI chat, dashboards, and the advisory hub.

Is my data safe in the Research Lab tools? Yes. Every Research Lab tool runs entirely in your browser using only the inputs you type. No data is uploaded, and there is no PHI handling — the tools use mock or user-provided figures only.

Are the tool outputs real predictions? No. Every tool is an illustrative planning model with clearly labeled assumptions, intended to teach trade-offs and support scenario planning — not to produce audited forecasts or vendor quotes.

The Book

Is the book free? The book is integrated into the platform and readable there; distribution terms are set by the owner.

Can I listen instead of read? Yes — every chapter has audio narration at `/book/listen` and transcripts in the reader.

How does the book connect to the tools? Each pillar chapter ends with a “Work This Chapter on the Platform” table linking to the specific tools that let you model the chapter’s argument; Appendix E lists all tools.

The Platform

Which tool should I use for X? Ask the AI Analyst — it maps questions to tools. Or consult the in-app guide at `/academy/getting-started/research-lab`.

Can I save my work? Bookmark any page/tool to My Library (`/saved`); use the Research Workspace to save and compare analysis scenarios and export findings.

Why can’t I access the AI chat / a dashboard? Those are gated to subscribers and above. Upgrade at `/upgrade`.

The Academy

How is the Academy different from the book? The book is the argument; the Academy is structured, lesson-by-lesson instruction with hands-on tool practice and credentials.

Do I get a certificate? Yes — completing a course issues a certificate, verifiable at /verify/[hash].

Billing

How do subscriptions work? Via Stripe — monthly or yearly, per tier (subscriber, student, professional, team). Manage billing in the customer portal under /account/billing.

What happens if my payment fails? A lifecycle email is sent; access reverts according to subscription status synced from Stripe.

Appendix G — Book Chapter ↔ Platform Tool Cross-Reference

This table summarizes which Research Lab tools each pillar chapter’s “Work This Chapter on the Platform” section links to. (The authoritative mapping lives in chapters.ts; this is a reading aid.)

Chapter (pillar)	Primary linked tools
Ch 1 — Framework & Execution Sequence	HTR Simulator; Transformation Friction Index
Ch 2–3 — Policy	Policy Simulator; Work Requirements Calculator; H.R. 1 Cliff Scenario
Ch 4 — Technology (Data Infrastructure)	FHIR Interoperability Lab; Risk Stratification Engine; EMR/EHR Lab; Statewide EHR Deployment Modeler
Ch 5 — Technology in Practice	Clinical Data Exchange Lab; AI Clinical Governance Lab; Digital Health Lab
Ch 6–7 — Economics	APM Design Lab; Shared Savings Calculator; Global Budget Transition Modeler; Hospital Financial Stress Test; CEA Calculator
Ch 8–9 — Clinical	Risk Stratification Engine/Methodology; VBC Quality Measures; High vs. Low Value Care; Clinical Quality Optimizer
Ch 10 — Equity	Health Equity Studio; Population Health Modeler
Ch 11 — Operations	Transformation Scorecard; Workforce Modeler; CIN & Shared Services Modeler; EMS Transformation Modeler
Ch 12–16 — Knowledge transfer, future, sustainability, portfolio, AHS	Innovation Leaderboard; VBC Readiness Assessment; Evidence Library; HTI Dashboard

Appendix H — API Surface Summary

The platform exposes ~39 API routes. The ones operators most often touch:

Endpoint	Purpose	Protection
/api/chat	Proxy to the AI backend	Tier-gated
/api/search	Site search	Public
/api/wire, /api/wire/comments	The Wire feed + comments	Public read / auth write
/api/stripe/checkout , /portal, /team-checkout	Billing flows	Auth
/api/stripe/webhook	Subscription sync → roles	Stripe signature
/api/academy/certificates	Certificate issue/verify	Auth
/api/personalized-learning (+ /audio)	Learning-path generation	Auth
/api/hospitals, /api/hti-scores, /api/state-metrics, /api/rht-states	Dashboard data	Public (some cached)
/api/keys/create rotate revoke	Developer API keys	Auth
/api/cron/digest, /api/digest	Email digests	CRON_SECRET/DIGEST_SECRET
/api/webhooks/sanity	Content change hooks	Signature
/api/health	Liveness	Public

Appendix I — Glossary of Vermont Programs & Entities

- **GMCB** — Green Mountain Care Board; Vermont’s health-care regulator (rate review, hospital budgets, global budgets).
- **AHS** — Vermont Agency of Human Services; the system operator for transformation.
- **DVHA** — Department of Vermont Health Access; Medicaid + HIE governance.
- **Blueprint for Health** — Vermont’s medical-home and community-health-team program.
- **VCCI** — Vermont Chronic Care Initiative; intensive case management for the highest-cost Medicaid members.
- **SASH** — Support and Services at Home; an aging-in-place program.
- **CCBHC** — Certified Community Behavioral Health Clinic.
- **CHT** — Community Health Team (Blueprint).
- **RHT Program** — the \$195M federal Rural Health Transformation award to Vermont (5 years, ending FY2030).
- **AHEAD** — the CMS all-payer global-budget model Vermont is adopting (performance year from Jan 2028).

- **Act 167 (2022)** — initiated Vermont’s reform cascade; mandated the statewide-EHR feasibility assessment.
- **Act 68 (2025)** — made transformation mandatory; the framework RHT capital executes.
- **OneCare** — Vermont’s former ACO, whose wind-down informs the dependency-sequence lessons.

Appendix J — Document Maintenance

This guide is a living document. Update it whenever the ecosystem changes:

- **A tool is added/removed** → update §13 bench table, §14 walkthroughs, Appendix A, Appendix G, and the tool count everywhere.
- **The book is revised** → update Part II, Appendix G, and any chapter references.
- **The architecture changes** → update Part V and the relevant ops sections.
- **Env vars change** → update §30 and Appendix C.
- **Launch state changes** → update Part VIII.

Regenerate the .docx from this .md (via pandoc) after edits. Keep the version line at the top current.

This completes the HTR Ecosystem Guide — a comprehensive user, technical, and operational manual spanning the book, the platform, and the Academy.